

Autumn Crossword

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ONE ACROSS

At an autumn festival, 35 identical maple leaves are divided among four labeled wreaths A, B, C, D . The numbers of leaves placed on A and B must each be congruent to 1 (mod 3) and cannot exceed 22. The numbers placed on C and D must each be divisible by 3 and cannot exceed 27. How many ordered quadruples (A, B, C, D) of leaf counts are possible?

ONE DOWN

A team of 8 friends is spending the weekend picking apples. They need to divide themselves into 3 non-empty groups to harvest the North, South, and East orchards simultaneously. However, two of the friends, Alice and Bob, have an ongoing rivalry about who the fastest picker is and refuse to be assigned to the same orchard. Let N be the number of distinct ways the 8 friends can be assigned to the 3 specific orchards. What is $N/10$?

TWO DOWN

A large wooden basket contains R red apples and G green apples. To test for ripeness, a farmer draws two apples at random from the basket without replacement. The probability that both apples drawn are the exact same color is exactly $\frac{1}{2}$. If the total number of apples in the basket is between 240 and 260, what is the maximum possible number of red apples?

THREE DOWN

During the grand autumn harvest, Farmer Appleseed numbers his trees sequentially starting from 1 up to 100. The number of apples on tree n , denoted as A_n , follows a peculiar pattern: the sum of the apples on any two consecutively numbered trees is exactly the square of the larger tree's number. That is, $A_n + A_{n-1} = n^2$ for all integers $n \geq 2$. If the first tree yields exactly 1 apple, what is the total number of apples harvested from the first 15 trees?

FOUR ACROSS

During an experiment, a quantity associated with a falling maple leaf is modeled by a cubic polynomial $P(x)$. It is known that $P(x+1) - P(x) = 3x^2 - 143x + 396$ for all real x , and that $P(2) = 256$. If the three roots of $P(x)$ are a, b , and c , find $(a+1)(b+1)(c+1)$.

FIVE ACROSS

A triangular banner at a fall festival is shaped like triangle ABC . The altitude from A meets BC at H , and M is the midpoint of BC . The incircle of triangle ABC is tangent to BC at D , with the points occurring in the order B, H, D, M, C . Suppose $AH = 24$, $AM = 6\sqrt{41}$, and $HD = 6$. Find the area of triangle ABC .